

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 7: Coordinate Geometry

Previous year board questions, topic-wise: 5 questions

The tag after each question gives the board paper and the marks. Solutions are in a separate PDF with the same question numbers.

1 The distance formula

1. AD is a median of $\triangle ABC$ with vertices $A(5, -6)$, $B(6, 4)$ and $C(0, 0)$. Length AD is equal to:
(a) $\sqrt{68}$ units (b) $2\sqrt{15}$ units (c) $\sqrt{101}$ units (d) 10 units **CBSE 2024, 1 mark**
2. If the distance between the points $(3, -5)$ and $(x, -5)$ is 15 units, then the values of x are: (a) 12, -18 (b) -12, 18 (c) 18, 5 (d) -9, -12 **CBSE 2024, 1 mark**

2 The section formula

3. The centre of a circle is at $(2, 0)$. If one end of a diameter is at $(6, 0)$, then the other end is at:
(a) $(0, 0)$ (b) $(4, 0)$ (c) $(-2, 0)$ (d) $(-6, 0)$ **CBSE 2024, 1 mark**
4. Find the ratio in which the point $\left(\frac{8}{5}, y\right)$ divides the line segment joining the points $(1, 2)$ and $(2, 3)$. Also, find the value of y . **CBSE 2024, 3 marks**
5. $ABCD$ is a rectangle formed by the points $A(-1, -1)$, $B(-1, 6)$, $C(3, 6)$ and $D(3, -1)$. P , Q , R and S are mid-points of sides AB , BC , CD and DA respectively. Show that diagonals of the quadrilateral $PQRS$ bisect each other. **CBSE 2024, 3 marks**